

Charanjeet Khullar

Data Engineer

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Work Authorization: US Citizen

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State: New Jersey

Professional experience:

- Overall 9+ years of experience as a Data Engineer and 12+ years in the IT industry on SQL, Spark, Hive, ETL, Data modeling (erwin), Snowflake, Hbase, and HDFS.
- Extensive experience in Data warehouse, Data modeling, Data integration, and Data migration using ETL jobs.
- Innovative database delivery/data warehousing solutions to finance industries with HA solutions.
- Proficient in handling Data modeling, System/Data analysis, Design and Development for both OLTP and Data warehousing environments.
- Acute knowledge on spark architecture and real-time streaming.
- Hands-on experience with configuration of Hadoop Ecosystem components: Hadoop (Horton Works), HDFS, MapReduce, Hlve and good knowledge on Impala, Kafka, Strom, Pig, Sqoop, Oozie, Hbase, Zookeeper, and Flume.
- Strong experience in test-driven development, complete application life cycle development
- Expertise in mapping business requirements into functional requirements.
- Working experience in code repositories like Github, Tortoise git & Gitlab.
- Best practices in Agile & Scrum methodologies.
- Good working knowledge of Amazon Web Services (AWS) Cloud Platform which includes services like EC2, S3, VPC, ELB, IAM, DynamoDB, Cloud Front, Cloud Watch, Route 53, Auto Scaling, Security Groups, EC2 Container Service (ECS), Code Commit, Code Pipeline, Code Build, Code Deploy, Dynamo DB, Auto Scaling, Security Groups, CloudWatch, CloudTrail, Ops Works, Kinesis, IAM, SQS.
- Experience in Data Analysis, Data Profiling, Data Integration, Migration, Data governance and Metadata Management, Master Data Management and Configuration Management.
- Experienced in building Automation Regressing Scripts for validation of ETL process between multiple databases like Oracle, SQL Server, Hive, and Mongo DB using Python
- Expert in building Enterprise Data Warehouse or Data warehouse appliances from Scratch using both Kimball and Inmon's Approach.

Technical Skills

Programming Language:	SQL, PL/SQL, and UNIX. Cassandra, Django
Big Data Tools:	Hadoop Ecosystem: Map Reduce, Spark 2.3, Airflow 1.10.8, Nifi 2, HBase 1.2, Hive 2.3, Pig 0.17 Sqoop 1.4, Kafka 1.0.1, Oozie 4.3, Hadoop 3.0, Apache NiFi
Database:	MYSQL, Snowflake, PostgreSQL, OracleDB, Hive, Hbase, MongoDB, DynamoDB
Operating System:	Windows, Unix, Sun Solaris
Data tool :	Erwin , Informatica, Tableau
Query language :	SQL, PL/SQL, and UNIX.Cassandra, Django
Cloud Platform :	AWS, Azure, Google Cloud.Snowflake
OLAP Tools:	Tableau, SSAS, Business Objects, andCrystal Reports 9, MongoDB
ETL/Data warehouse Tools:	Informatica 9.6/9.1, and Tableau.

Highest Education

MCA (**Master's in Computer Application**) from India completed in September 2011. Education Credential **Evaluated degree from ECE** has the United States Equivalent (Credit Score **(3.4/4.0)** Masters In Computer Science.

B.Com (Pass) (**Graduation degree in Commerce**) from India completed in October 2008. Education Credential **Evaluated degree from ECE** has the United States equivalent.

Work experience:

Sr. Data engineer,
Employer: Samach LLC.

July 2015 - Present

Client: Crown Media Group, NY

NOV 2023 – Present

Role and Responsibility:

- **Writing Spark Jobs for Data Consumption:** Responsible for writing Spark jobs to read data from S3, transforming it as per business requirements, and ingesting it into HDFS and Redshift. This involves ensuring data integrity and efficiency in the transformation process.
 - **Building an Enterprise Crown Media:** Involves aggregating data from the HDFS raw zone to create a unified, abstracted view for business units. This task supports advanced analytics and requires deep knowledge of business logic and data processing.
 - **Scheduling Batch Spark Jobs via Oozie Scheduler:** Responsible for automating and scheduling batch Spark jobs using the Oozie scheduler, ensuring that jobs are run on a predefined frequency with minimal errors.
 - **Writing Shell Scripts for Data Cleansing:** Develop shell scripts to cleanse data from various sources, ensuring that data is formatted and filtered correctly before processing.
 - **Following Agile Methodologies:** Adhering to Agile principles, participating in sprints, and collaborating with the team to ensure project deadlines and milestones are met effectively.
 - **End-to-end handling of Projects/Application Requests:** Taking ownership of data engineering projects from initiation to delivery, managing timelines, addressing technical challenges, and ensuring alignment with business goals and requests.
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- **Environment:** Spark, Redshift, Python, HDFS, Hive, Pig, Sqoop, Python, Kafka, Shell scripting, Linux, Jenkins, Eclipse, Git, Oozie, Talend, Agile Methodology.

Data Engineer

Aug 2022 - July 2023

Client: Samsung, NYC

Role and Responsibility:

- Designed and implemented data pipelines using AWS Glue components (crawlers, jobs, triggers).
- Configured AWS Glue to integrate with databases, data lakes, and streaming services.
- Created AWS Glue jobs for data transformations like cleaning, enrichment, and aggregation.
- Developed ETL workflows with scheduling and triggers for automated job execution.
- Monitored pipeline performance and troubleshooted issues like job failures and data quality.
- Implemented security measures, including data encryption and access control.
- Optimized and scaled the pipeline for performance and increasing data volumes.
- Documented pipeline architecture and collaborated with teams to align with business goals.
- Automated data backups to S3 using AWS CLI and migrated applications to AWS Glue and Athena.

- Implemented data warehouse solutions with AWS Redshift, migrating data from on-premise sources.
- Developed ETL solutions and database architecture using MS SQL Server, Oracle, and others.
- Built and maintained data pipelines with Python and Bash scripts on Unix/Linux systems.
- Implemented CI/CD processes using Jenkins for automated deployment.
- Used PySpark for data aggregation, cleansing, and transformation in AWS Glue.
- Applied predictive analytics and statistical models for business insights.
- Worked with Big Data tools like Spark, Kafka, HBase, and AWS Glue for data processing.
- Automated test scripts and scheduled jobs with Oozie workflows.
- **Environment:** Spark, Redshift, Python, HDFS, Hive, Pig, Sqoop, Python, Kafka, Shell scripting, Linux, Jenkins, Eclipse, Git, Oozie, Talend, Agile Methodology.

Data Engineer

Employer: Bergen County, New Jersey, NJ

July 2015 - May 2022

Role and Responsibility:

- **Data Ingestion Pipeline Development:** Developed and maintained data ingestion pipelines using Talend ETL tools and Bash scripting, integrating big data technologies such as Hive, Impala, Spark, Kafka, and Talend for efficient data processing.
- **Secure Python Pipelines:** Designed secure and scalable data pipelines in Python for handling large datasets, ensuring data security and integrity throughout the process.
- **Data Source Requirement Gathering:** Collected and defined the requirements for integrating new data sources, including lifecycle management, data quality checks, transformations, and metadata enrichment.
- **Data Quality Assurance:** Implemented robust data quality checks within the data pipelines to ensure data accuracy, completeness, and consistency.
- **Data Engineering Support:** Provided support for data engineers by exploring data, performing ad-hoc ingestions, and offering expertise to data scientists using big data technologies.
- **Machine Learning Model Building:** Built machine learning models using PySpark and MLlib, showcasing big data capabilities for predictive analytics and decision-making.
- **Data Ingestion Framework Enhancement:** Improved and strengthened the data ingestion framework, creating more secure and robust data pipelines for handling large-scale data efficiently.
- **Real-Time Data Streaming:** Enabled data streaming capabilities by implementing Kafka and Talend integration for multiple data sources, enabling real-time data processing.
- **Storage and Database Integration:** Managed and worked with various storage formats (Avro, Parquet) and databases (Hive, Impala, Kudu), optimizing data retrieval and storage processes.
- **S3 Data Lake Management:** Administered S3 Data Lake management, handling inbound and outbound data requests and ensuring smooth data operations in the big data platform.
- **Cluster Security Knowledge:** Applied knowledge of security components like Kerberos, Sentry, SSL/TLS for securing data processing and ensuring encrypted communications across systems.

- **Agile Data Modeling:** Contributed to developing agile, flexible data modeling patterns, ensuring the data architecture could easily adapt to changing requirements.
- **Job Automation via JILs:** Implemented JILs (Job Information Language) to automate job execution processes within the production cluster, enhancing operational efficiency and workflow automation.
- **Troubleshooting Bugs:** Addressed and resolved user-reported issues related to data analysis, using JIRA and IRIS for tracking and ensuring efficient bug resolution.
- **SCRUM Methodology Participation:** Collaborated with the SCRUM team to deliver user stories and tasks within agreed timelines for every sprint, adhering to agile principles.
- **Resolving Production Job Failures:** Identified and resolved production job failures in multiple scenarios, ensuring the reliability and stability of the data pipeline.
- **UNIX Scripting for Automation:** Developed UNIX shell scripts to define data workflows, process data files, and automate tasks within the pipeline, reducing manual effort.
- **Automation in Production Cluster:** Automated jobs within the production cluster using JILs, improving job execution scheduling and reducing manual intervention.
- **Technical Environment Expertise:** Proficient in working with technologies including Spark, Redshift, Python, HDFS, Hive, Pig, Sqoop, Kafka, Shell scripting, Linux, Jenkins, Eclipse, Git, Oozie, Talend, and Agile methodology.

Python engineer

Employer: Bluethink

Jan 2013 - Dec 2014

- **Twitter data analysis:** Using Web Scraper to retrieve the twitter data of top 50 popular users by using API and Developed Pig Latin scripts where extensive coding needs to be reduced to analyze large data sets.
- **YouTube data analysis:** Retrieved categories in which most numbers of videos are uploaded, rated and viewed
- **Telecom data:** Analyzed one million users of telecom data (network carrier) and offered premium plans for valuable customers resulted in 15% revenue increase for the company